EXP_2026Q2_CART_NUDGES · PRE-REGISTERED DESIGN BRIEF · ENGINEER

GROWTH · CART · SEALED 2026-06-05T15:42:42Z · DURATION 14 DAYS

# Cart-page upsell nudges

*Adding a 'Frequently bought with items in your cart' module on the cart page increases average order value through additional complementary purchases.*

AT A GLANCE

**Primary metric is *conversion_rate*; decision is binary against Ship if AOV CI excludes zero on benefit side.**

*Pre-registered before any analysis. Decision is binary against conversion_rate: a 2.0% relative relative lift over a 10.0% baseline requires 2,500 per arm at the standard α / power settings. Guardrails sit on revenue_per_user.*

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SAMPLE REQUIRED (PER ARM)

**2,500**

*to detect a 2.0% relative relative effect at α = 0.05, power = 0.80*

BASELINE

10.0%

TOTAL SAMPLE

5,000

DURATION

14 days

FEASIBILITY

VIABLE

MECHANISM

## Why we think this works

*Cart visitors have high intent. A complementary-product nudge at this moment converts more often than the same module on the PDP, based on the 2025Q3 recs experiment.*

*Cart sessions are only ~2k/day, which is the binding constraint on this test. AOV baseline ~$48. We knew this would be a tight powered experiment going in.*

METRICS

## conversion_rate is the verdict-driving metric.

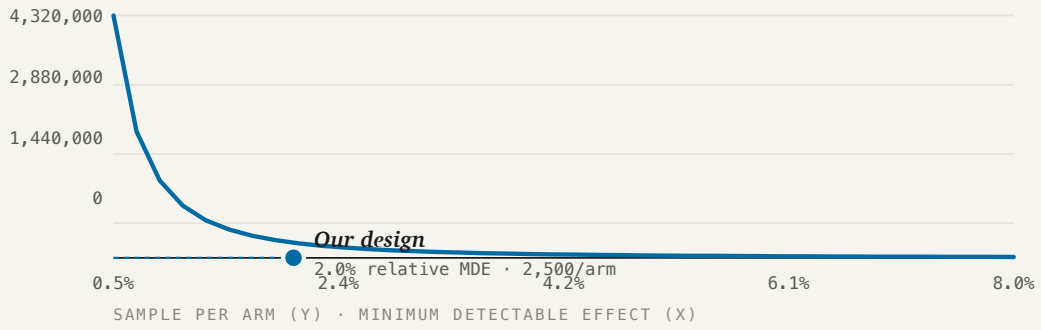
*Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.*

| | |
|---|---|
| PRIMARY | **conversion_rate**<br>Proportion · baseline 10.0% · MDE 2.0% relative · direction higher is better |
| SECONDARY | **conversion_rate**<br>Leading indicator. Not used for the ship/no-ship decision but informs interpretation. |
| GUARDRAIL | **revenue_per_user**<br>Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative. |

POWER

## Detecting smaller effects requires much more sample.

*To resolve a 2.0% relative relative effect against a 10.0% baseline, we need 2,500 per arm. Tighter detection thresholds would climb the curve to the left — quickly.*




*Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.*

ASSIGNMENT + COHORTS

## First-exposure attribution at the user_id level.

*Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.*

| | |
|---|---|
| UNIT | **user_id**<br>One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed. |
| ARMS | **50 / 50**<br>control · treatment |
| COHORT | **all assigned users** |

DECISION RULES

## Pre-registered. The analyze verb walks the same tree against the same metrics.

*Ship if AOV CI excludes zero on benefit side. Likely to be underpowered — be prepared for an inconclusive outcome and a longer-running follow-up.*

| | |
|---|---|
| SHIP | Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7 |
| LIFT WITH CAVEAT | Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out |
| NO-SHIP · GUARDRAIL | Any guardrail's 90% CI lies past its harm threshold |
| NO-LIFT | Well-powered, primary CI straddles zero with half-width > 2 × MDE |
| INCONCLUSIVE | Underpowered (n < required) AND primary CI straddles zero |
| INVALID | SRM $\chi^2$ $p < 0.0005$ and no override resolved — investigate assignment |

INTEGRITY LOCK · THREE-PART SEAL

# brief.sealed.yaml binds intent + metric definitions + expected shape.

*Tampering with any of the three components after seal will fail* `agentxp.schemas.brief_seal.verify_or_raise`*;* `agentxp analyze` *refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.*

```
DESIGN_CHAIN_HASH
bf842f93af13d54f6357c50cbb17d0431fb25010b8442e7f1d041f512c606857
METRIC_SNAPSHOT         1 metric hashed
CONVERSION_RATE
fbf7a228872bb052817eb550deb3311acba51c3cb60f03241c460211b307f2d0
EXPECTED_SHAPE          user_id · 50 / 50
SEALED_AT               2026-06-05T15:42:42Z
SEALED_BY               walk_experiment_demo
```

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**BRIEF.SEALED.YAML** SEAL BF842F93AF13… · 2026-06-05T15:42:42Z · R1 R11